VAN DINH NGHIA

FPGA DESIGN INTERN

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ABOUT ME

Electronics and Telecommunications Engineering student focused on FPGA design and AI hardware acceleration. Experienced in Verilog HDL, Xilinx Vivado, RTL verification, synthesis, timing analysis, INT8 quantization, AXI4-Lite integration, and board-level AI inference. Interested in building efficient, verifiable FPGA accelerators for edge AI systems.

EDUCATION

Ho Chi Minh City University of Technology and Education — **Expected Graduation: 2027**

B.Eng. in Electronics and Telecommunications Engineering — GPA: 8.43/10 (3.50/4.00)

Specialization: Telecommunications and Integrated Circuit Design. Relevant coursework: Digital Logic Design, Verilog HDL, FPGA Design, Computer Architecture, Digital Signal Processing, and Embedded Systems.

TECHNICAL SKILLS

HDL	Verilog HDL, RTL Design, FSM, Pipelining, INT8 Quantization, Fixed-Point Requantization
FPGA Tools	Xilinx Vivado, Simulation, Synthesis, Implementation, Timing and Resource Analysis
Interfaces	UART, AXI4-Lite, Memory-Mapped I/O, PS–PL Integration
Programming	Embedded C, C/C++, Python, MATLAB, Linux/ARM64, Git
Platforms	Xilinx KV260, ZedBoard, Tang Nano

RESEARCH EXPERIENCE

INT8 1D-CNN AI Inference Accelerator on Xilinx KV260 — **01/2026 – 08/2026**

FPGA/SoC AI Hardware Accelerator

- Deployed an end-to-end 5-class 1D-CNN inference pipeline, mapping convolution, ReLU, and max-pooling to integer-only INT8 FPGA logic while executing GAP/Dense classification on the ARM processor.
- Converted floating-point inference using per-layer scale/zero-point parameters and fixed-point requantization for efficient INT8 hardware execution.
- Designed the accelerator with 20 MAC lanes, 5 requantization units, 42 execution contexts, AXI4-Lite control, and configurable on-chip memories.
- Achieved bit-exact RTL validation: 10,240/10,240 outputs matched golden results across 8 samples; all predictions matched the TFLite model.
- Met 100 MHz timing with +0.701 ns WNS and 21,437-cycle inference latency; placed design used 17,530 LUTs (14.97%), 6,898 FFs (2.94%), and 0 DSP/BRAM.
- Integrated the accelerator with the KV260 PS through AXI4-Lite; developed a Linux ARM64 C application and achieved 98.31% board-level accuracy (14,756/15,009 samples correct).

FPGA Multiplier Architecture Optimization — **10/2025 – 03/2026**

Student Scientific Research Project — Verilog RTL

- Designed, simulated, synthesized, and implemented nine signed 16-bit multipliers: DSP, array, shift-add, 2/8-stage pipeline, Booth radix-2/radix-4, and 2/8-core architectures.
- Achieved 100% pass rate across 114 functional test cases; deployed and verified selected designs on Xilinx KV260, ZedBoard, and Tang Nano boards through a 115200-baud UART link with automated C-based PASS/FAIL checking.
- Reached 725.16 MHz with the 8-stage pipeline versus 108.57 MHz for the array baseline (nearly 7× higher Fmax); Booth radix-4 reduced latency from 16 to 8 cycles.
- Demonstrated linear multi-core throughput scaling, with the 8-core architecture producing eight results per eight cycles (one result/cycle).

HONORS & ACTIVITIES

Teaching Assistant — Digital Design and Programming Laboratories

- Guided students in combinational/sequential logic, simulation, source-code debugging, and practical laboratory assignments.
- Assisted lecturers in preparing laboratory materials and supporting hands-on engineering sessions.

Principal Investigator — Student Scientific Research Project

- Led the project "Analysis and Experimental Evaluation of Algorithm Optimization Techniques on FPGA" from RTL development through multi-board validation and final technical reporting.

LANGUAGES

Vietnamese: Native　　**English:** Technical documentation, reading, and communication